Supplementary Material for "Cardinality Coverage in Evolutionary Feature Selection: A Budget-Aware Study of Initialization and Search Operators"

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The tables below give the effect sizes and per-dataset results that the paper summarizes. Sections S1 to S4 list the datasets, give the implementation details, examine missing runs and the choice of classifier, and show two figures of the pilot. They are generated by `make_tables.py` from the CSV files in the paper's repository, which, like the experiment repository with the distilled run logs, will be made public after review. Group names, configurations, and measures are those of the paper; OBL is the opposition-based start and MIIP our implementation of NSGAII-MIIP.

Table S1: Principal paired comparisons in the benchmark, datasets as units: mean difference, 95% bootstrap interval over datasets, matched-pairs rank-biserial correlation r, and Holm-corrected Wilcoxon p over the three pairs of configurations within a group and measure. This family differs from that of Table 4 of the paper (four groups), so the reach comparison of the two starts has slightly different corrected values there. Time to reach: restricted mean share of the budget spent before the archive holds a subset with at most ten features. Largest subset: number of features of the largest subset on the returned front

	Decision tree				Random forest				LightGBM			
Comparison	Mean	95% CI	r	p	Mean	95% CI	r	p	Mean	95% CI	r	p
Reach, sparse − OBL	+0.680	[+0.507, +0.840]	+1.00	<0.001	+0.635	[+0.471, +0.782]	+1.00	<0.001	+0.543	[+0.364, +0.721]	+1.00	<0.001
Reach, MIIP − OBL	+0.043	[+0.007, +0.090]	+1.00	0.125	+0.075	[+0.019, +0.150]	+1.00	0.016	+0.081	[+0.015, +0.170]	+1.00	0.062
Reach, sparse − MIIP	+0.641	[+0.469, +0.800]	+1.00	<0.001	+0.537	[+0.369, +0.700]	+1.00	<0.001	+0.464	[+0.286, +0.643]	+1.00	<0.001
AUF_{10}, sparse − OBL	+0.491	[+0.374, +0.601]	+0.94	<0.001	+0.477	[+0.361, +0.586]	+0.93	<0.001	+0.415	[+0.290, +0.539]	+0.85	<0.001
AUF_{10}, MIIP − OBL	+0.058	[+0.015, +0.112]	+0.87	0.005	+0.057	[+0.007, +0.111]	+0.59	0.031	+0.083	[+0.025, +0.150]	+0.62	0.029
AUF_{10}, sparse − MIIP	+0.437	[+0.315, +0.556]	+0.89	<0.001	+0.393	[+0.273, +0.511]	+0.95	<0.001	+0.332	[+0.202, +0.465]	+0.64	0.005
AUF_{20}, sparse − OBL	+0.482	[+0.361, +0.598]	+0.94	<0.001	+0.465	[+0.344, +0.579]	+0.92	<0.001	+0.402	[+0.273, +0.533]	+0.83	<0.001
AUF_{d}, sparse − OBL	+0.140	[+0.096, +0.183]	+0.89	<0.001	+0.127	[+0.080, +0.174]	+0.82	<0.001	+0.085	[+0.046, +0.126]	+0.66	0.006
Best F1, sparse − OBL	−0.007	[−0.018, +0.007]	−0.58	0.013	−0.029	[−0.043, −0.016]	−0.76	<0.001	−0.022	[−0.032, −0.013]	−0.87	<0.001
Best F1, sparse − MIIP	−0.012	[−0.028, +0.002]	−0.51	0.036	−0.021	[−0.037, −0.006]	−0.61	0.004	−0.021	[−0.036, −0.007]	−0.67	0.003
Largest subset, sparse − OBL	−348.7	[−591.3, −157.5]	−0.99	<0.001	−2632.6	[−6927.3, −225.5]	−0.99	<0.001	−120.6	[−189.0, −61.3]	−0.98	<0.001
Time to reach, sparse − OBL	−0.771	[−0.888, −0.634]	−1.00	<0.001	−0.778	[−0.881, −0.658]	−1.00	<0.001	−0.724	[−0.847, −0.587]	−1.00	<0.001
Time to reach, MIIP − OBL	−0.044	[−0.081, −0.015]	−1.00	0.002	−0.065	[−0.110, −0.030]	−1.00	<0.001	−0.070	[−0.114, −0.033]	−1.00	<0.001

Table S2: Split of the AUF difference (sparse minus OBL start, $K = 10$) into reach and quality parts (Eq. 2 of the paper), by group. Mean over datasets of Δ, ρ, and κ; signed share S and absolute share A (Eq. 3), in percent with 95% bootstrap intervals over datasets; the share of the net difference that comes from limits the OBL start never reaches on a dataset; and the numbers of datasets with a negative and a positive quality part

Group	Data	$\bar{\Delta}$	$\bar{\rho}$	$\bar{\kappa}$	S	A	OBL never	$\kappa < 0/\kappa > 0$
Decision tree (designed)	8	+0.623	+0.624	−0.001	100.1 [99.9, 100.5]	99.8 [99.4, 100.0]	99.2 [97.0, 100.5]	1/1
Decision tree	30	+0.491	+0.491	+0.000	100.0 [99.8, 100.2]	99.7 [99.4, 99.9]	93.8 [85.8, 99.4]	5/5
Random forest	34	+0.477	+0.483	−0.006	101.2 [99.5, 104.3]	98.1 [95.2, 99.7]	93.1 [83.8, 100.2]	6/7
LightGBM	28	+0.415	+0.415	+0.000	100.0 [99.3, 100.8]	99.2 [98.3, 99.8]	88.4 [74.6, 97.4]	6/7

Table S3: Matched-size difference in test F1, sparse minus OBL start, for every dataset on which it is defined. $|M|$: number of limits up to ten that both starts reach in at least three of five seeds; m: matched-size difference; seed-paired: number of seeds with a shared reached limit, mean, minimum, and maximum of the per-seed differences. Datasets with at most 50 test rows are left out only in the exploratory analysis

						Seed-paired			
Group	Dataset	d	Test rows	$\|M\|$	m	Seeds	Mean	Min	Max
Decision tree (designed)	satimage	36	1286	10	−0.007	5	−0.007	−0.009	−0.003
Decision tree	PenDigits	16	2199	10	−0.000	5	−0.000	−0.002	+0.002
Decision tree	ItalyPowerDemand	24	219	10	+0.004	5	+0.004	+0.000	+0.018
Decision tree	Crop	46	4800	10	+0.004	5	+0.004	−0.003	+0.009
Decision tree	ProximalPhalanxOutlineCorrect	80	178	10	−0.009	5	−0.009	−0.024	+0.014
Decision tree	TwoLeadECG	82	232	2	−0.009	5	−0.007	−0.047	+0.035
Decision tree	Libras	90	75	10	−0.001	5	−0.001	−0.050	+0.055
Decision tree	ElectricDevices	96	3327	8	+0.005	5	+0.008	−0.009	+0.022
Decision tree	MedicalImages	99	229	10	−0.009	5	−0.009	−0.025	+0.016
Decision tree	ECG5000	140	1000	6	+0.002	4	+0.003	−0.008	+0.010
Decision tree	Adiac	176	156	9	+0.004	4	−0.008	−0.055	+0.011
Random forest	PenDigits	16	2199	10	−0.001	5	−0.001	−0.002	+0.000
Random forest	ItalyPowerDemand	24	219	10	−0.001	5	−0.001	−0.007	+0.009
Random forest	Crop	46	4800	10	+0.000	5	+0.000	−0.003	+0.004
Random forest	ProximalPhalanxOutlineCorrect	80	178	10	+0.001	5	+0.001	−0.015	+0.023
Random forest	TwoLeadECG	82	232	6	+0.022	5	+0.021	+0.011	+0.027
Random forest	Libras	90	75	10	+0.011	5	+0.011	−0.013	+0.068
Random forest	ElectricDevices	96	3327	3	+0.008	5	+0.008	−0.004	+0.022
Random forest	MedicalImages	99	229	10	+0.007	5	+0.007	−0.006	+0.016
Random forest	ECG5000	140	1000	6	+0.001	4	+0.001	−0.004	+0.005
Random forest	Adiac	176	156	8	+0.028	4	+0.021	−0.017	+0.054
Random forest	Strawberry	235	196	9	−0.009	4	−0.015	−0.024	−0.005
Random forest	Coffee	286	11	10	−0.138	5	−0.138	−0.258	+0.053
Random forest	BeetleFly	512	8	10	−0.106	5	−0.106	−0.494	+0.254
LightGBM	PenDigits	16	2199	10	+0.000	5	+0.000	−0.000	+0.000
LightGBM	ItalyPowerDemand	24	219	10	+0.005	5	+0.005	−0.006	+0.020
LightGBM	Crop	46	4800	10	−0.000	5	−0.000	−0.004	+0.004
LightGBM	ProximalPhalanxOutlineCorrect	80	178	10	−0.002	5	−0.002	−0.013	+0.011
LightGBM	TwoLeadECG	82	232	5	+0.004	5	+0.003	−0.022	+0.020
LightGBM	Libras	90	75	10	−0.007	5	−0.007	−0.042	+0.022
LightGBM	ElectricDevices	96	3327	4	+0.007	5	+0.008	+0.000	+0.016
LightGBM	MedicalImages	99	229	10	−0.001	5	−0.001	−0.011	+0.007
LightGBM	ECG5000	140	1000	6	+0.001	4	+0.000	−0.005	+0.010
LightGBM	Adiac	176	156	4	+0.036	4	+0.025	−0.042	+0.081
LightGBM	Strawberry	235	196	10	−0.005	5	−0.005	−0.016	+0.020
LightGBM	Coffee	286	11	10	+0.000	5	+0.000	−0.091	+0.091
LightGBM	BeetleFly	512	8	10	−0.029	5	−0.029	−0.267	+0.140

Table S4: Finished runs out of planned runs, by group and configuration

Group	Configuration	Finished	Planned	Share
Decision tree (designed)	OBL start	40	40	100%
Decision tree (designed)	Sparse start	40	40	100%
Decision tree	OBL start	164	195	84%
Decision tree	Sparse start	165	195	85%
Decision tree	NSGAII-MIIP	149	195	76%
Random forest	OBL start	170	195	87%
Random forest	Sparse start	195	195	100%
Random forest	NSGAII-MIIP	160	195	82%
LightGBM	OBL start	134	195	69%
LightGBM	Sparse start	194	195	99%
LightGBM	NSGAII-MIIP	133	195	68%

Table S5: Number of features at which half of the runs reach ten features, from the logistic model of reach on $\log_2 d$ and $\log_2$ of the budget, at the median budget of 6,000 fits; 95% percentile interval over 2,000 bootstrap resamples of datasets (resamples in which all runs agree are skipped). Descriptive locations for these datasets and budgets

Group	Configuration	d_{50}	95% CI	Resamples
Decision tree (designed)	OBL start	172	[92, 174]	1826
Decision tree	OBL start	163	[122, 202]	2000
Decision tree	NSGAII-MIIP	208	[159, 253]	2000
Random forest	OBL start	212	[145, 277]	2000
Random forest	NSGAII-MIIP	268	[224, 343]	2000
LightGBM	OBL start	226	[148, 300]	2000
LightGBM	NSGAII-MIIP	266	[210, 302]	2000

Table S6: Pilot: fits until the archive of an OBL run with ten times the budget first holds a subset with at most ten features; median and range over five seeds, in multiples of the benchmark budget, and per feature

Dataset	d	Budget	Median fits	Range	Budgets	Fits per feature
Adiac	176	12,000	6,534	6,128–8,452	0.54	37
Strawberry	235	6,000	9,292	5,472–16,662	1.55	40
Earthquakes	512	12,000	47,830	46,324–60,611	3.99	93
Computers	720	12,000	85,239	74,518–87,064	7.10	118

S1. Datasets

Table S7 lists the 46 datasets of the designed experiment and the benchmark, which Table 2 of the paper summarizes. The four pilot datasets are Adiac, Strawberry, Earthquakes, and Computers from this list.

Table S7: Datasets of the designed experiment and the benchmark (Table 2 of the paper), ordered by the number of features d. Train and test give rows; the validation part has as many rows as the test part. Budget: model fits per run

Dataset	Domain	d	Train	Test	Classes	Budget
PenDigits	handwriting	16	6,594	2,199	10	2,000
ItalyPowerDemand	electricity	24	658	219	2	2,000
satimage[a]	satellite	36	3,858	1,286	6	6,000
Crop	satellite	46	14,400	4,800	24	4,000
ProximalPhalanx-OutlineCorrect	image outline	80	535	178	2	12,000
TwoLeadECG	ECG	82	698	232	2	2,000
Libras	gesture	90	210	75	15	12,000
ElectricDevices	electricity	96	9,983	3,327	7	6,000
MedicalImages	image	99	683	229	10	12,000
ECG5000	ECG	140	3,000	1,000	5	6,000
Wafer	industrial sensor	152	4,300	1,432	2	2,000
ChlorineConcentration	simulation	166	2,585	861	3	4,000
Adiac	image outline	176	469	156	37	12,000
EOData[a]	satellite	182	29,976	10,008	24	6,000 / 2,000
LSST	astronomy	216	2,955	985	14	6,000
Strawberry	spectra	235	591	196	2	6,000
USPS[a]	handwriting	256	5,578	1,860	10	6,000
Coffee	spectra	286	34	11	2	12,000
Ham	spectra	431	128	43	2	12,000
FordA	engine sensor	500	2,953	984	2	6,000
BeetleFly	image outline	512	24	8	2	12,000
Earthquakes	seismic	512	275	93	2	12,000
har[a]	motion sensor	561	6,181	2,059	6	6,000
OliveOil	spectra	570	36	12	4	2,000
BasicMotions	motion sensor	600	48	16	4	2,000
Epilepsy	motion sensor	618	163	56	4	12,000
Computers	electricity	720	300	100	2	12,000
UWaveGestureLibraryAll	gesture	945	2,686	896	8	6,000
StarLightCurves	astronomy	1,024	5,542	1,847	3	2,000
COIL20[a]	object images	1,024	880	280	20	6,000
NATOPS	motion capture	1,224	216	72	6	12,000
FingerMovements	EEG	1,400	248	84	2	6,000
CinCECGTorso	ECG	1,639	852	284	4	6,000
PhonemeSpectra	speech	2,387	4,016	1,326	39	2,000
HandOutlines	image outline	2,709	822	274	2	12,000
gisette[a]	handwriting	5,000	4,200	1,400	2	6,000
FaultDetectionA	vibration	5,120	8,184	2,728	3	6,000
EthanolConcentration	spectra	5,253	316	104	4	12,000
SelfRegulationSCP1	EEG	5,376	337	112	2	12,000
Prostate_GE[a]	gene expression	5,966	62	20	2	6,000
FaceDetection	MEG	8,928	5,650	1,882	2	12,000
arcene[a]	mass spectrometry	10,000	120	40	2	12,000
Heartbeat	heart sound	24,705	245	82	2	12,000
EigenWorms	motion	107,904	155	52	5	12,000
PEMS-SF	traffic	138,672	264	88	7	2,000
MotorImagery	ECoG	192,000	226	76	2	6,000

[a] Dataset of the experiment designed for this comparison. EOData is also in the benchmark; its two budgets are those of the designed experiment and of the benchmark.

S2. Implementation and Reproducibility

This section gives the details summarized in Section 4.6 of the paper. The experiment repository provides the code, data loaders, fixed-seed splits, and run logs with fronts, generation archives, fit counts, stopping causes, and timings. The paper repository contains the derived tables, analysis scripts, checksums, and this supplementary material. Both will be made public after review.

Time series are flattened channel by channel without scaling; rows with missing values are removed. The decision-tree experiments ran on a workstation, and the random forest and LightGBM on eight cores of a computing cluster. Every run log records the code commit of both repositories and refuses uncommitted code, except the random-forest runs: the cluster nodes had no git, and their commits were reconstructed from the history of the cluster clone. Library versions are not stored per run. The experiment records give Python 3.11, NumPy 2, and scikit-learn 1.9.0 on the workstation and 1.9.1 on the cluster; the LightGBM version is not recorded. A reproduction guide in the paper repository lists the commits, split settings, and commands for every number.

S3. Missing Runs and the Choice of Classifier

Missing runs. Of the 195 planned benchmark runs per configuration and classifier, 69 to 87% finished for the OBL start, 85 to 100% for the sparse start, and 68 to 82% for NSGAII-MIIP (Table S4); all 80 runs of the designed experiment finished. Under the decision tree, 47 benchmark runs reached the six-hour limit (6 OBL, 30 sparse, and 11 NSGAII-MIIP runs), under LightGBM 64 (36, 1, and 27), and under the random forest none. The combinations that were not run remove 25 OBL and 35 NSGAII-MIIP runs from each group. Missing runs are thus concentrated in the configurations and on the wide datasets where reach is low. Figure S1 repeats the main comparison under the three treatments of Section 4.3 of the paper. In every group and under every treatment, the OBL start never reaches the limit in more seeds than the sparse start, and the absolute reach share A is at least 97.9%. These treatments are scenarios; they do not rule out that runs that did not finish would have changed the comparison in another way. Counting every missing run as reaching nothing moves the mean difference in reach from 0.68, 0.64, and 0.54 to 0.59, 0.68, and 0.67. One decision outside the main comparison changes: with only complete datasets, the AUF difference between NSGAII-MIIP and the opposition-based start under LightGBM has Holm $p = 0.055$ instead of 0.029.

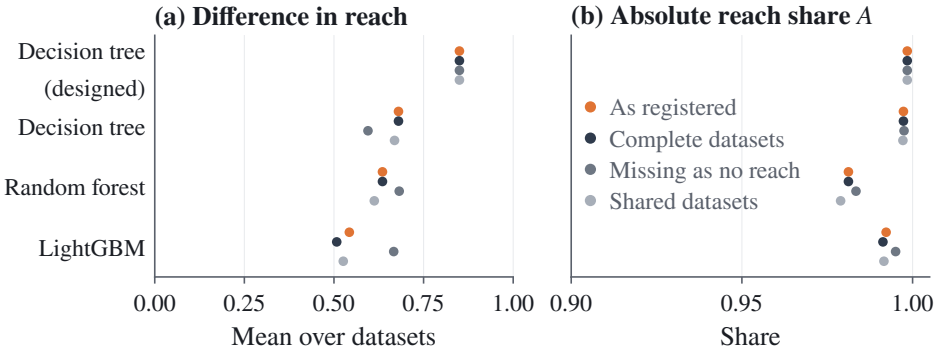


Figure S1: The main comparison persists under the tested treatments of missing runs. (a) Mean difference in reach, sparse minus OBL start; (b) the absolute share A of the AUF difference due to reach. As registered: finished runs only; complete datasets: datasets on which all seeds finished; missing as no reach: missing runs count as reaching nothing; shared datasets: datasets on which every configuration has runs

The classifier. On the 26 datasets on which both starts have runs under all three classifiers, the AUF lead of the sparse start is 0.465 under the decision tree, 0.462 under the random forest, and 0.448 under LightGBM (Fig. S2). The Friedman test gives $p = 0.007$, but only the difference between the tree and the forest remains after Holm correction ($+0.003$, $p = 0.022$), and the absolute reach share A is 99.7%, 99.4%, and 99.5%. The share of opposition-based runs that reach ten features on its 27 common datasets is 0.36, 0.42, and 0.44 (Friedman $p = 0.082$). The classifier changes the scores more than the comparison: the sparse start's own AUF is 0.04 lower under the tree than under the other two (Holm $p < 0.001$).

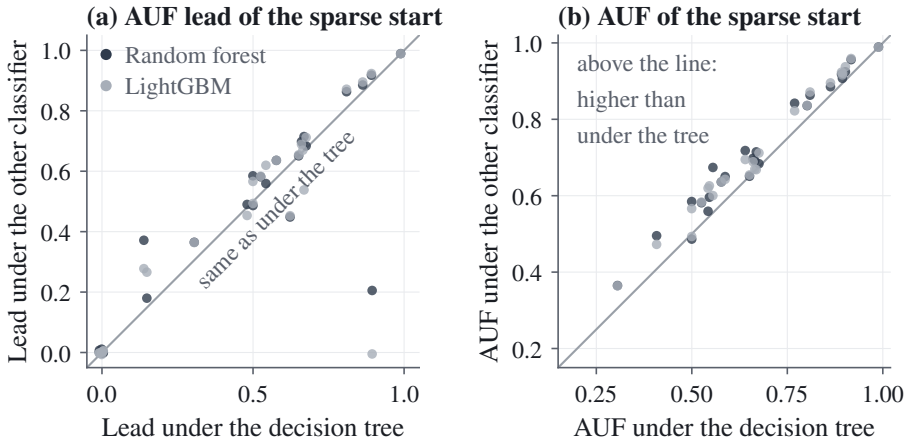


Figure S2: The classifier changes the scores but hardly the comparison (Section S3). Each point is one of 26 datasets, under the random forest or LightGBM against the decision tree. (a) The AUF lead of the sparse start over the opposition-based start stays close to the diagonal. (b) The sparse start's own AUF lies above it: both other classifiers score higher than the tree.

S4. Pilot: Unchanged Children and the Forced Flip

The two figures belong to Section 5.7 of the paper, which reports their numbers.

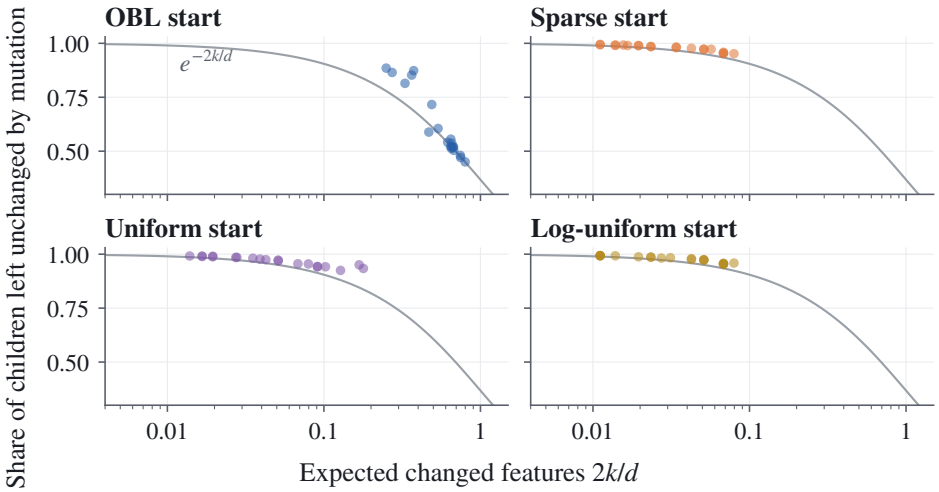


Figure S3: Balanced mutation leaves children unchanged at about the rate the analysis predicts. One point per run with balanced mutation: the share of children that mutation left unchanged, against $2k/d$ at the run's median size k of newly evaluated subsets; the curve is $e^{-2k/d}$ (Section 3.5 of the paper). An approximate check: the exact prediction would average $e^{-2k/d}$ over the sizes of all mutated children, which were not logged

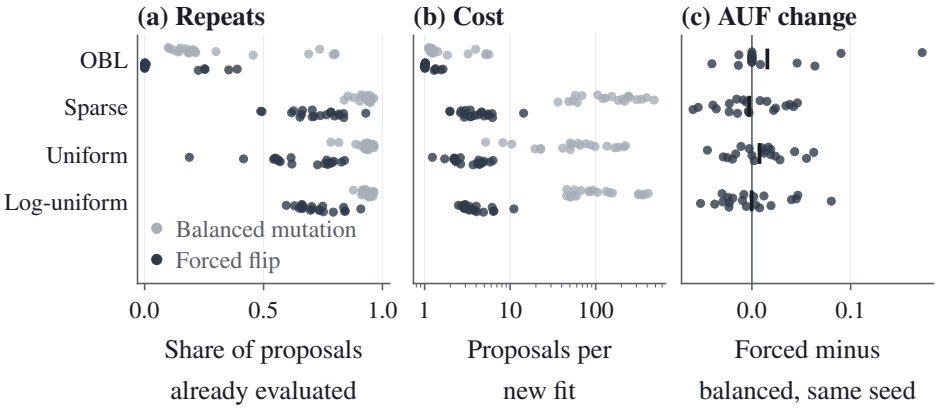


Figure S4: A forced flip makes the small-subset searches far cheaper but not better. Per start, one point per run: (a) the share of proposed subsets that had already been evaluated, (b) proposals per new model fit (logarithmic), (c) AUF with the forced flip minus AUF with balanced mutation, same seed and initial population; bars are means